



PROJECT REPORT

MINI-PROJECT 2: ENGLISH ↔ PREDICATE LOGIC TRANSLATOR

'DEVELOPING A PYTHON PROGRAM
FOR LOGICAL CONVERSION'

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DEPT: COMPUTER ENGINEERING

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GRADE: 5 MARKS



1. What the project does (Description):

This Python program simplifies the connection between natural English sentences and first-order predicate logic. It allows users to:

- *Translate simple quantitative English sentences (such as all, some, none) into their corresponding formal logical symbols.*
- *Convert predicate logic expressions into understandable English sentences.*
- *Interact with a command-line interface to select the translation direction.*

2. How the program works:

The program was built using Python's re (regular expressions) library for pattern matching. To convert English to logic: The program scans the beginning of a sentence for keywords such as "all," "some," or "not." It then uses regular expressions to extract the subject and predicate. If "all" is detected, it applies the total quantifier (for all x) and the implicative (right arrow). If "some" is detected, it applies the existential quantifier (exists x) and the conjunction (wedge). To convert the logic to English: The program looks for logical symbols ($\forall$, $\exists$, $\neg$). Once a pattern is matched, it extracts the terms within the predicates and reformats them into Standard English syntax.

3. Examples of Inputs and Outputs:

<i>Input Type</i>	<i>Input Example</i>	<i>Output Result</i>
<i>English to Logic</i>	<i>All cats are cute</i>	$\forall x(cat(x) \rightarrow cute(x))$
<i>English to Logic</i>	<i>Some students are smart</i>	$\exists x(students(x) \Delta smart(x))$
<i>Logic to English</i>	$\forall x(dog(x) \rightarrow friendly(x))$	<i>All dogs are friendly</i>
<i>Logic to English</i>	$\neg \exists x(fish(x) \Delta fly(x))$	<i>No fishs are fly</i>